Kyucheol Min

Cambridge, MA | mink202@mit.edu

RESEARCH INTERESTS

Generative modeling and neural sampling; ML for molecular and cellular systems; statistical mechanics

EDUCATION

MASSACHUSETTS INSTITUTE OF TECHNOLOGY (MIT) Cambridge, MA

Bachelor of Science Expected May 2027

Double Major in **Electrical Engineering and Computer Science (6-2)** and **Physics (8)**

Cumulative GPA: 5.0/5.0

Highlighted coursework: Machine Learning (Grad), Deep Learning (Grad), Discrete Probability and Stochastic Processes (Grad), Algorithms for Inference (Grad, in progress), Quantum Computation (Grad), Electromagnetics (Grad), Quantum Physics II, Statistical Physics, Fundamentals of Statistics, Large-Scale Inference (in progress), Computer Architecture

PUBLICATIONS

- **K. Min**, E. Hofgard, T. Smidt. "FrOGS: Discrete Neural Sampler for Independent Alloy Configurations Across Chemical Conditions." *NeurIPS AI4Mat Workshop* (**Spotlight**), 2026. arXiv:2609.02948. Multicomponent extension in preparation for *npj Computational Materials*.

- E. Hofgard, **K. Min**, N. Segal, D. W. Mittan-Moreau, A. M. Tehrani, V. Oklejas, R. Gómez-Bombarelli, J. Nigam, D. W. Paley, A. S. Brewster, T. Smidt. "Learning Lattice Parameters from Powder X-Ray Diffraction Data Using Invariants." *Acta Crystallographica Section A* (accepted), 2026. arXiv:2607.21829

RESEARCH EXPERIENCE

FrOGS: Discrete Neural Sampler Across Chemical Conditions Mar 2026 – Present

Undergraduate Researcher (Project Lead) at Atomic Architects Lab | PI: **Tess Smidt**

- Conceived and developed a discrete neural sampler that couples an autoregressive model to a learned continuous-time Markov chain, generating independent configurations from the Boltzmann distribution across chemical conditions
- Matched exact free energies of the 2D Ising model on a 20×20 lattice to within 0.05% across its critical temperature
- Recovered all 3 CuAu ordered phases, whereas a published neural baseline missed $CuAu_3$, a thermodynamically delicate phase just below the convex hull of the free energy
- Used as a baseline by researchers at Microsoft Research and the University of Cambridge (He et al., arXiv:2610.01739)

Representation Learning of Single-Cell Images Jul 2026 – Present

Undergraduate Researcher (Project Lead) at Uhler Lab | PI: **Caroline Uhler**

- Developed a self-supervised framework that separates perturbation-driven morphology from plate-to-plate technical variation; trained on 4.3M cells across 3,030 CRISPR gene knockouts
- Improved cross-plate replicate retrieval ~7× over DINOv2 and ~9× over Cell-DINO (mAP 0.0036 and 0.0027 → 0.0249) on held-out genes, and raised the share of knockouts above AUC 0.6 (22% vs 5–6% for the baselines)
- Confirmed the metrics were consistent on held-out genes from a function-disjoint split, suggesting generalization

Learning Lattice Parameters from Powder X-Ray Diffraction Data Using Invariants Sep 2024 – Jun 2026

Undergraduate Researcher at Atomic Architects Lab | PI: **Tess Smidt**

- Designed a transformer encoder architecture and trained it on the MP-20 dataset for bispectrum (an invariant lattice geometry descriptor) prediction from XRD patterns
- Showed that bispectrum intermediate representations outperform direct prediction for lattice parameter reconstruction from XRD data, reducing the mean absolute percentage error of cell lengths by ~2× on the experimental RRUFF dataset

Pulse Sequence Compiler for FPGA-Controlled Superconducting Qubit Experiments Jun 2025 – Sep 2026

Undergraduate Researcher (Project Lead) at Quantum Coherent Electronics Lab | PI: **Kevin O'Brien**

- Developed a pulse sequence compiler integrating Fermilab's Quantum Instrumentation Control Kit (QICK) into MIT's internal Python quantum framework (QCML), translating multi-channel experiment definitions and parameter sweeps into FPGA-native programs for qubit control and readout
- Demonstrated an end-to-end superconducting qubit characterization with the compiler, including T_1 and T_2 measurements
- Adopted by a team of 4 researchers at MIT Lincoln Lab for further development including multiplexing and active reset

HONORS

54th International Chemistry Olympiad (IChO) Gold Medal: Ranked 10th of 326 participants Jul 2022
Talent Award of Korea: Awarded by the Korean Deputy Prime Minister and Minister of Education Dec 2022
Kwanjeong Scholarship: Full-tuition scholarship awarded to ~3 South Korean undergraduates per year 2023 – Present
Eta Kappa Nu (HKN): IEEE Honor Society for Electrical Engineering and Computer Science 2025

TALKS & PRESENTATIONS

Winter AI for Materials & Chemistry (AI4MC) Workshop, Poster Dec 2026
Spotlight talk, NeurIPS AI4Mat Workshop Dec 2026

TEACHING & SERVICE

Mentor of students in introductory electromagnetics (8.021)
Co-founder of SIGMAS, the undergraduate statistics society

TECHNICAL SKILLS

Languages: Python, C/C++, RISC-V Assembly
ML & Scientific Computing: PyTorch, pymatgen, e3nn, QuTiP, QICK
Tools: Git, Linux, SLURM